JACKSON ESHBAUGH

+1 484.484.3326 | eshbaugj@lafayette.edu | jacksoneshbaugh.github.io

jacksoneshbaugh | jacksoneshbaugh | jacksoneshbaugh

Bethlehem, PA, USA

Research Interests

Computational semantics; computational linguistics; natural language processing; idioms and multiword expressions; neural network interpretability

Education

- **Lafayette College** — *August 2023–Expected May 2027*
  B.S., Computer Science; A.B., French — Easton, PA
  - GPA: 4.00/4.00
  - Minor: Mathematics

Honors and Fellowships

- **Beinecke Scholar** — *2026*
- **Phi Beta Kappa** — *Inducted 2026*
  Lafayette College
- **President, Upsilon Pi Epsilon** — *Inducted 2026; President, 2026–Present*
  Lafayette College
- **EXCEL Scholar** — *May 2025–Present*
  Lafayette College Department of Computer Science
- **Whitman Fellow** — *January 2025*
  Lafayette College Department of Economics
- **Marquis Scholarship** — *May 2023*
  Lafayette College
- **Dean's List** — *All Semesters*
  Lafayette College

Research Experience

- **Honors Thesis Research—Computational Linguistics** — *February 2025–Present*
  Department of Computer Science, Lafayette College — Easton, PA
  - Advisor: Dr. Sofia Serrano
  - Formalizing linguistic theories of idiomaticity into computational frameworks for multilingual NLP.
  - Operationalizing dimensions such as semantic opacity, syntactic flexibility, and conventionality for model design and evaluation.
  - Investigating how language models represent idiomatic meaning through interpretability techniques.
- **Independent Study—Linguistics of Idioms** — *January 2026–May 2026*
  Department of Language and Literary Studies, Lafayette College — Easton, PA
  - Advisor: Dr. Maria Hernandez
  - Examined foundational and contemporary theories of idiomatic language, including structural, construction-based, and cognitive approaches.
  - Analyzed the limitations of structural linguistic models in explaining idiomatic meaning and behavior.
  - Authored a French-language research paper examining the implications of modern linguistic theory for future computational models of idiomaticity.

- **Fully Independent Research—Neural Network Interpretability** *April 2025–Present*
  Department of Computer Science, Lafayette College Easton, PA
  - Investigated when linear surrogate models fail to faithfully represent neural networks.
  - Designed an experimental framework comparing surrogate fidelity and predictive accuracy across regression tasks.
  - Demonstrated that high surrogate fidelity does not necessarily imply predictive accuracy and introduced diagnostic measures to characterize this discrepancy.
  - Implemented end-to-end machine learning pipelines for neural network training, surrogate evaluation, and statistical analysis.
  - Resulted in an accepted workshop paper at ICLR 2026 and an expanded conference manuscript currently in preparation.
- **EXCEL Scholar—Applied Machine Learning for Energy Systems** *May 2025–Present*
  Department of Computer Science, Lafayette College Easton, PA
  - Advisor: Dr. Jorge Silveyra
  - Conduct research on machine learning methods for residential building energy modeling, retrofit analysis, and neighborhood-scale energy planning.
  - Develop multimodal generative pipelines and EnergyPlus simulation workflows for synthetic residential building energy datasets.
  - First author of a submission on synthetic residential building energy generation (arXiv:2509.09794).
  - Presented work at KINBERCON 2025, Lafayette College EXCEL Scholars Poster Sessions, and Georgia State University.
  - Preparing manuscripts on calibration-first supervision without ground truth and graph-based algorithms for residential retrofit recommendation.
- **Whitman Fellow—Policy-Oriented Machine Learning** *January 2025*
  Departments of Economics & Computer Science, Lafayette College Easton, PA
  - Advisors: Dr. Gladstone Hutchinson and Dr. Frank Xia
  - Contributed to development of a structured county-level dataset for policy-oriented machine learning research.
  - Collected, cleaned, normalized, and integrated heterogeneous public datasets into a shared research database.
  - Supported downstream modeling efforts analyzing regional socioeconomic indicators.

Publications

[1] **Eshbaugh, J.** "Faithful to What?" On the Limits of Fidelity-Based Explanations. 2026. *Workshop on Scientific Methods for Understanding Deep Learning* at the *International Conference on Learning Representations*. arXiv:2506.12176.

[2] **Eshbaugh, J.**, Tiwari, C., and Silveyra, J. Synthetic Homes: An Accessible Multimodal Pipeline for Producing Residential Building Data with Generative AI. 2026. *Machine Learning with Applications*, Volume 25.

Presentations

[1] **Eshbaugh, J.** (Apr 2026). "Faithful to What? On the Limits of Fidelity-Based Explanations." *Workshop on Scientific Methods for Understanding Deep Learning, International Conference on Learning Representations 2026*, Rio de Janeiro, Brazil. [poster]

[2] Silveyra, J. and **Eshbaugh, J.** (Mar 2026). "From Images to Energy Models: Generative AI for Urban Energy Data." Georgia State University, Atlanta, GA, USA. [invited talk; slides]

[3] **Eshbaugh, J.** (Feb 2026). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Lafayette College EXCEL Scholars Poster Session*, Easton, PA. [poster]

[4] **Eshbaugh, J.** (Dec 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Lafayette College EXCEL Scholars Poster Session*, Easton, PA. [poster]

[5] **Eshbaugh, J.** (Oct 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *CyberAccelerate Poster Session at KINBERCON 2025*, Lancaster, PA. [poster]

[6] **Eshbaugh, J.** (Sep 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Lafayette College Bicentennial Weekend Poster Session*, Easton, PA. [poster]

TEACHING EXPERIENCE

- **Teaching Assistant** *August 2024–Present*
  Department of Computer Science, Lafayette College Easton, PA
  - Support instruction during class meetings by assisting students with programming, debugging, and conceptual problem solving.
  - Provide individualized help with course material in introductory computer science contexts.
  - Recognized by students for clear explanations, accessibility, and care for student learning.
- **Mentored Study Group Leader Supporting Computer Science** *August 2024–Present*
  Academic Resource Hub, Lafayette College Easton, PA
  - Lead two weekly collaborative review sessions to reinforce course material and support student learning.
  - Design worksheets, review activities, and exam-preparation materials.
  - Create a welcoming learning environment focused on conceptual understanding and peer-supported practice.

PROFESSIONAL SERVICE

- **Association for Computational Linguistics (ACL) Rolling Review** *October 2025–November 2025*
  Secondary Reviewer with Dr. Sofia Serrano
  - Evaluated manuscript methodology, experimental design, and validity of scientific claims.
  - Drafted preliminary reviews refined through discussion with Dr. Serrano.
- **Machine Learning with Applications** *April 2026–Present*
  Reviewer

EMPLOYMENT

- **Campus Tour Guide / Admissions Ambassador** *September 2023–June 2025*
  Admissions Office, Lafayette College Easton, PA
  - Led campus tours for prospective students and families, presenting academic programs and student life through structured public presentations.
  - Represented Lafayette College to visitors and supported recruitment efforts through direct engagement with applicants.
- **IT Specialist Intern** *June 2024–August 2024*
  WFMZ-TV Allentown, PA
  - Supported broadcast operations through installation, troubleshooting, and maintenance of newsroom and studio technology.
  - Assisted with teleprompter systems, workstation upkeep, and field deployment of technical equipment.

LEADERSHIP AND SERVICE

- **Lafayette DiscipleMakers Christian Fellowship** *September 2023–Present*
  President; student leadership and audio/visual production
  - President, 2026–Present.
  - Selected to Oversight Team (2025) and Shepherding Team (2024), student leadership bodies responsible for organizational direction, decision-making, and ministry care.
  - Provide A/V production support for large-group gatherings.
- **Audio/Visual Specialist** *July 2021–Present*
  Hope Alliance Church, Bethlehem, PA
  - Support weekly worship services through projection, livestream, and technical production.

Additional Information

Languages: English (native); French (fluent, professional working proficiency)

Technical Skills: Python, Java, C; LaTeX; EnergyPlus; TensorFlow; PyTorch; data analysis and visualization workflows

Creative and Media Production: Trombone; jazz vocals; piano; composition; photography and photo editing; videography and video editing; live A/V production